EXCEL

Excel for Data Analysts

The functions that cover daily analyst work, plus the 2 debugging questions every interviewer asks.

- **8 functions with real examples**
- **The #N/A checklist**
- **Why SUM shows 0**

Lookup and summarise

This is 70% of Excel work in any analyst job.

- **XLOOKUP**

  Looks left or right, handles "not found" on its own. Replaces VLOOKUP.

  ```
  =XLOOKUP(A2, Emp[ID], Emp[Name], "Not found")
  ```

- **SUMIFS**

  Sum with any number of conditions.

  ```
  =SUMIFS(Sales, Region, "North", Month, "Mar")
  ```

- **COUNTIFS**

  Count with conditions. Great for open tickets per owner.

  ```
  =COUNTIFS(Status, "Open", Owner, G2)
  ```

- **IFS**

  Clean buckets without nesting 5 IFs.

  ```
  =IFS(B2>=90,"A", B2>=75,"B", TRUE,"C")
  ```

Clean and shape

Raw data is never clean. These save hours every week.

- **TRIM + CLEAN**

  Removes extra and invisible spaces that break lookups.

  ```
  =TRIM(CLEAN(A2))
  ```

- **TEXTSPLIT**

  Split "Delhi, North, 2026" into 3 columns.

  ```
  =TEXTSPLIT(A2, ", ")
  ```

- **UNIQUE + FILTER**

  Live lists that update when data changes.

  ```
  =FILTER(Data, Data[Sales] > 50000)
  ```

- **TEXT**

  Turn dates into readable labels for reports.

  ```
  =TEXT(A2, "mmm-yy")
  ```

VLOOKUP shows #N/A

Interviewers ask this to see if you have worked with real data. Check in this order.

- **1. Hidden spaces**

  "A101 " and "A101" do not match. Wrap the lookup value.

  ```
  =VLOOKUP(TRIM(A2), Range, 2, FALSE)
  ```

- **2. Number stored as text**

  One side is 101, the other is "101". Green triangle in the corner gives it away. Fix with VALUE or Text to Columns.

- **3. Range not locked**

  Range shifts when copied down. Use $A$2:$C$500 or a Table.

- **4. Missing FALSE**

  Without it Excel does an approximate match.

SUM shows 0

The numbers are there, the total is 0. Three causes, in order of how often I see them.

- **Numbers stored as text**

  Left aligned, green triangle. Select the column, Data, Text to Columns, Finish.

- **Calculation is Manual**

  Someone switched it to speed up a big file. Formulas, Calculation Options, Automatic.

- **Circular reference**

  The SUM range includes its own cell. Status bar shows the cell address.

Interview tip: say the checks in order. It shows you have debugged real files, not just practised clean data.